

Alessandro Simoni, Ph.D.

Deep Learning Research Engineer

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Links: 🏠 alexj94.github.io | 💼 LinkedIn | 🐙 GitHub | 🎓 Google Scholar

Summary

I am a Deep Learning Research Engineer with 2+ years of experience developing AI solutions for real-world industrial applications. My work has covered image reconstruction, diffusion models, semantic segmentation, 3D human pose estimation, and text-to-audio models. I earned a Ph.D. in Deep Learning and Computer Vision from University of Modena and Reggio Emilia in April 2024, focusing on 3D human-centric scene understanding and 3D object reconstruction. I am passionate about research and contributing to advancements in the field of Deep Learning.

Work Experience

Covision Lab	April 2024 – (Now)
Deep Learning Research Engineer	Brixen, Italy
• Activities: Diffusion Models, Text-to-Audio Models, Discrete Tomography, Semantic Segmentation, Classification	
AlmageLab (University of Modena and Reggio Emilia)	May 2020 – April 2024
Research Fellow / PhD Student	Modena, Italy
• Activities: 3D Object Reconstruction (neural rendering, NeRF), 3D Skeleton Pose Estimation	

Education

University of Modena and Reggio Emilia	November 2020 – April 2024
Ph.D. in Deep Learning and Computer Vision	Modena, Italy
• Thesis: <i>From Images to 3D Space: The Role of Semantic Keypoints for 3D Perception</i>	
University of Modena and Reggio Emilia	September 2017 – February 2020
MSc in Computer Science	Modena, Italy
University of Modena and Reggio Emilia	September 2013 – February 2017
BSc in Computer Science	Modena, Italy

Skills

Spoken languages: Italian (mothertongue), English (C1 certified)
Programming languages: Python (proficiency), C/C++ (basic)
Libraries & Tools: PyTorch, OpenCV, Numpy, TensorRT, VSCode, PyCharm
Deep Learning: 3D Pose Estimation, 3D Reconstruction, Image Diffusion Models, Text-to-Audio Models, Image Reconstruction, Semantic Segmentation, Image Classification

Relevant Publications (* = equal contribution, full list on Google Scholar)

Bounding Box-Guided Diffusion for Synthesizing Industrial Images and Segmentation Map	2025
A. Simoni *, F. Pelosin *	CVPRW
3D Pose Nowcasting: Forecast the future to improve the present	2025
A. Simoni *, F. Marchetti *, G. Borghi, F. Becattini, L. Seidenari, R. Vezzani, A. Del Bimbo	CVIU
Dynamic Label Injection for Imbalanced Industrial Defect Segmentation	2024
E. Caruso *, F. Pelosin *, A. Simoni *, M. Boschetti	ECCVW
KRONC: Keypoint-based Robust Camera Optimization for 3D Car Reconstruction	2024
D. Di Nucci *, A. Simoni *, M. Tomei, L. Ciuffreda, R. Vezzani, R. Cucchiara	ECCVW

Depth-Based Privileged Information for Boosting 3D Human Pose Estimation on RGB	2024
A. Simoni, F. Marchetti, G. Borghi, F. Becattini, D. Davoli, L. Garattoni, G. Francesca, L. Seidenari, R. Vezzani	ECCVW
Depth-based 3D human pose pose refinement: Evaluating the refinet framework	2023
A. D'Eusanio, A. Simoni, S. Pini, G. Borghi, R. Vezzani, R. Cucchiara	PRL
Semi-Perspective Decoupled Heatmaps for 3D Robot Pose Estimation from Depth Maps	2022
A. Simoni, S. Pini, G. Borghi, R. Vezzani	IROS + RA-L
Multi-category mesh reconstruction from image collections	2021
A. Simoni *, S. Pini *, R. Vezzani, R. Cucchiara	3DV
Future urban scene generation through vehicle synthesis	2021
A. Simoni, L. Bergamini, A. Palazzi, S. Calderara, R. Cucchiara	ICPR

Certifications

First Certificate in English (FCE)

December 2020